Every finite characteristic-three field: trace and geometric torsion

Result. For every finite characteristic-three field and nonsingular general Weierstrass model, Lean proves the exact source gcd(q+1−#E,q)≠1 predicate iff three divides the actual trace iff actual point count is one modulo three iff b2=0 iff j=0 iff actual geometric three-torsion count is one iff its ZMod3 rank is zero. The source predicate is invariant under finite field extensions. No field-size/degree bound or Hasse/trace-square/Endrank premise. [pcc:unverified]

Every finite char3	Source gcd(trace,q)≠1 ↔ 3 divides trace ↔ #E≡1 mod3 ↔ b2=0 ↔ j=0 ↔ geometric three-torsion count 1
Actual proof route	Complete square; finite cube root; paired twist fibers Ordinary: one partner has a 3-point; a=0: neither does
Explicit controls	Ordinary F3 pair: #E=5/3, rational 3-torsion=1/3 Source-SS F3 pair: #E=7/1, traces −3/+3

Ordinary branch. For $y^2=x^3+a\cdot x^2+b\cdot x+c$ with $a\neq 0$, finite cube-root surjectivity supplies an actual third-division-polynomial abscissa. Complementary square-root fibers supply an ordinate on the curve or its nonsquare-scaled partner. One partner therefore has an actual nonzero point killed by three. Actual Cauchy/Lagrange and twist-source invariance exclude the source predicate. No rational-torsion-existence or non-isomorphism premise. [pcc:unverified]

Special branch and general models. If a=0, the actual division polynomial excludes nonzero three-torsion on both partners. Cauchy makes both point counts nonzero modulo three; their actual sum2(q+1), with $q=3^n$ and n>0, forces each count to be one modulo three. A complete-square unit-u1 shift preserves discriminant, actual additive point group/cardinality and b2, and its x^2 coefficient equals original b2. It transfers the result to every general model. Prior actual group/geometric j/module proofs are credited. [pcc:unverified]

Controls. The credited ordinary F3 curve $y^2=x^3+2x^2+1$ has j=1, five actual points and rational three-torsion count one. Its d=2 partner has three points, rational three-torsion count three and P=(1,1) of exact order three; both source predicates are false. Lean negates the rational replacement criterion. The credited source-SS F3 curve $y^2=x^3+2x+1$ and its partner have seven/one points and traces−3/+3. Lean directly refutes the universal trace-zero strengthening. The source p≥5 prime-field clause is retained. [pcc:unverified]

Primary review. FST2010 selectedpp230/237: exact trace predicate/qualifications. Galbraith Ex9.5.4 count context, preceding inspectedPDF7 and actual cubic-with-x^2 proof credited; the false non-isomorphism exercise is not a premise. Milne Remark3.16/contextpp98–99/p75notes: geometric p-rank/Endrank distinction. Official finite Frobenius/cardinality/coprime-power and preceding actual division-polynomial/Cauchy/geometric j proofs reviewed. KB/Conductor unavailable, bounded fallback before code. No fullread/corpus promotion/novelty. [pcc:unverified]

Validation and scope. Whole gate 55 declarations, zero failures, zero trusted native exceptions. All 51 build/provenance files retained losslessly in 20 objects; five gate files, complete statement and recursive source/config pins. Lean4.35.0/Mathlib56a00585…, standard foundations only. Exact local PCC replay differs from signed standing. General p≥5 trace/geometric identifications, Endrank equivalences, Hasse/embedding/full-torsion-field bounds and remaining original fourteen-item gaps stay open. No holes/custom axioms/native evaluator/raised limits/new global flags/waivers/native experiment/paid compute/agent upload. [pcc:unverified]